



STA 4233/6233 – R Programming for Data Science

Final Project Report

**Data Exploration and Cleaning Report for ABC
Logistics Shipment Data**

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Table of Contents

Content Number	Title Of the Content	Page Number
1	Project Overview	3
2	Data Overview	3
3	Data Preparation and Variable Creation	4
4	Data Cleaning Process	5
5	Descriptive Statistics	8
6	Graphical Exploration	10
7	Correlation Analysis	14
8	Conclusion	16

1. Project Overview

ABC is a major technology company that manufactures and ships desktop and laptop products across North America and Latin America. The company operates with multiple manufacturing sites, multiple shipping modes, and warehouse locations. Because of recent supply chain disruptions and uncertainty in global transportation, ABC experienced significant difficulty in predicting shipment arrival times and managing customer expectations.

The primary objective of this project was to prepare and explore shipment data related to logistics movement from manufacturing sites to warehouses. The analysis focused on understanding shipment behavior, identifying data quality issues, calculating important lead time variables, and preparing a cleaned dataset for future modeling and business analysis.

This project specifically focused on data preparation and exploratory analysis rather than predictive modeling. The analysis included importing the shipment data, creating Quarter and Year variables using the calendar data, calculating Manufacturing Lead Time and In-transit Lead Time, cleaning missing and unusual values, exploring the variables using summary statistics and graphical methods, and examining relationships between variables through correlation analysis.

The final result of this project was a cleaned and structured dataset that can support future analysis of how shipment origin sites and shipping methods affect logistics lead time performance.

2. Data Overview

The project used data stored in the RawData.xlsx workbook. Two worksheets from the workbook were imported into R for analysis:

- Raw-Data
- Calendar

The Raw-Data worksheet contained shipment information for products transported from manufacturing sites to warehouses. The dataset included variables such as:

- LOB
- Origin
- Ship Mode
- PO Download Date
- Ship Date
- Receipt Date

The Calendar worksheet contained date ranges associated with Quarter and Year values. These date ranges were used to create additional time-related variables for the shipment records.

The original Raw-Data worksheet contained 9,124 rows and 6 columns before any data cleaning or preparation steps were performed.

The dataset was initially explored using summary functions, structure checks, missing value analysis, and category frequency tables. This exploratory process helped identify missing values, inconsistent categorical values, invalid lead times, and unusual observations that required cleaning before analysis.

Table 1 summarizes the number of rows received, removed, and retained for future analysis.

Table 1. Dataset Row Counts

Description	Count
Original rows received	9124
Rows removed during cleaning	145
Rows used for future modeling	8979

The dataset row counts demonstrated that only rows containing mostly missing values were removed during cleaning. The majority of observations were retained for future analysis.

3. Data Preparation and Variable Creation

Several data preparation steps were performed before the cleaning process began.

First, the date variables were converted into proper date formats in R to allow date calculations and comparisons. The variables converted into date format included:

- PO Download Date
- Ship Date
- Receipt Date

The Calendar worksheet was then used to create Quarter and Year variables for each shipment record. The Receipt Date variable from the Raw-Data worksheet was matched to the date ranges listed in the Calendar worksheet. This process allowed Quarter and Year values to be added dynamically without hardcoding date values.

Two important lead time variables were created during the preparation process.

Manufacturing Lead Time

Manufacturing Lead Time was calculated as:

Manufacturing Lead Time = Ship Date – PO Download Date

This variable represented the number of days required to manufacture and prepare the product before shipment.

In-transit Lead Time

In-transit Lead Time was calculated as:

In-transit Lead Time = Receipt Date – Ship Date

This variable represented the transportation time required for a shipment to move from the manufacturing site to the warehouse.

Table 2 summarizes the missing values identified immediately after the lead time variables were created.

Table 2. Missing Values After Lead Time Creation

lob	origin	ship_mode	po_download_date	ship_date	receipt_date	quarter	year	in_transit_lead_time	manufacturing_lead_time
0	0	0	0	145	145	145	145	145	145

The creation of these variables allowed the project to explore shipment performance and transportation efficiency across manufacturing sites and shipping methods.

4. Data Cleaning Process

The dataset contained several missing values, inconsistent categorical values, and unusual observations that required cleaning before analysis.

Missing Value Analysis

The original dataset contained missing values in the Ship Date and Receipt Date columns. Because these variables were used to calculate lead times, missing dates also created missing values in Manufacturing Lead Time and In-transit Lead Time.

The project instructions specified that date variables did not require direct cleaning or imputation. Therefore, the date columns themselves were not modified. Instead, the lead time variables created from those dates were cleaned and imputed appropriately.

Rows that contained mostly missing values were removed from the dataset. This process followed the project instructions, which stated that only rows containing mostly missing values should be deleted.

Table 3 summarizes the missing values identified in the original dataset.

Table 3. Original Missing Values

LOB	Origin	Ship Mode	PO Download Date	Ship Date	Receipt Date
0	0	0	0	145	145

Categorical Variable Exploration Before Cleaning

Frequency tables were created before cleaning to identify unusual values, inconsistent formatting, and category spelling problems.

Table 4 presents the LOB categories before cleaning.

Table 4. LOB Categories Before Cleaning

lob	n
Product B	5915
Product A	2947
Product C	262

Table 5 presents the Origin categories before cleaning.

Table 5. Origin Categories Before Cleaning

origin	n
Site A	3467
Site D	2434
Site C	1653
Site B	1570

Table 6 presents the Ship Mode categories before cleaning.

Table 6. Ship Mode Categories Before Cleaning

ship mode	n
AIR	3816
GROUND	2434
OCEAN	2344
FASTBOAT	530

Table 7 presents the Quarter categories before cleaning.

Table 7. Quarter Categories Before Cleaning

quarter	n
Q1	3862
Q2	3629
Q3	1445
NA	145
Q4	43

Cleaning of Categorical Variables

The categorical variables were cleaned to ensure consistency and remove formatting issues.

The following cleaning steps were performed:

- Extra spaces were removed
- Text values were converted into uppercase format

- Inconsistent category spellings were standardized
- Blank or unusual category values were corrected where necessary

The Ship Mode variable contained the category FASTBOAT, while the project description identified the shipping mode as FAST BOAT. Therefore, FASTBOAT was standardized to FAST BOAT to maintain consistency with the project requirements.

The categorical variables cleaned during the process included:

- LOB
- Origin
- Ship Mode
- Quarter

Cleaning of Lead Time Variables

The lead time variables required significant cleaning because they contained invalid and unrealistic values.

Negative In-transit Lead Time values were considered impossible because shipments cannot be received before they are shipped. Similarly, negative Manufacturing Lead Time values were treated as invalid because products cannot be shipped before the PO Download Date.

These invalid negative values were converted into missing values before imputation.

Outliers were also examined using the Interquartile Range (IQR) method. Extremely large or unrealistic lead time values outside the acceptable IQR range were treated as unusual observations and converted into missing values.

After invalid and unusual values were identified, missing lead time values were imputed using grouped median values based on Origin and Ship Mode combinations. If grouped median values were unavailable, the overall median lead time value was used as a fallback imputation method. The cleaning process produced a final dataset with consistent categorical variables, cleaned lead time variables, and substantially fewer missing values.

Table 8 summarizes the missing values remaining after the cleaning process.

Table 8. Final Missing Values

lob	origin	ship_mode	po_download	ship_date	receipt_date	quarter	year	in_transit_lead_time	manufacturing_lead_time
0	0	0	0	0	0	0	0	0	0

Final Category Tables After Cleaning

Table 9 presents the final LOB categories after cleaning.

Table 9. LOB Categories After Cleaning

lob	n
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PRODUCT B	5823
PRODUCT A	2901
PRODUCT C	255

Table 10 presents the final Origin categories after cleaning.

Table 10. Origin Categories After Cleaning

origin	n
SITE A	3412
SITE D	2394
SITE C	1627
SITE B	1546

Table 11 presents the final Ship Mode categories after cleaning.

Table 11. Ship Mode Categories After Cleaning

ship_mode	n
AIR	3752
GROUND	2394
OCEAN	2310
FAST BOAT	523

Table 12 presents the final Quarter categories after cleaning.

Table 12. Quarter Categories After Cleaning

quarter	n
Q1	3862
Q2	3629
Q3	1445
Q4	43

5. Descriptive Statistics

Descriptive statistics were calculated for both numeric and categorical variables included in the analysis.

Numeric Variables

The primary numeric variables included:

- Manufacturing Lead Time
- In-transit Lead Time

Summary statistics calculated for these variables included:

- Minimum
- First Quartile
- Median
- Mean
- Third Quartile
- Maximum
- Standard Deviation

These statistics helped summarize the center, spread, and variability of shipment lead times across the dataset.

Table 13 presents descriptive statistics for the numeric variables included in the analysis.

Table 13. Numeric Descriptive Statistics

in_transit_min	in_transit_q1	in_transit_median	in_transit_mean	in_transit_q3	in_transit_max	in_transit_sd	manufacturing_min	manufacturing_q1	manufacturing_median	manufacturing_mean	manufacturing_q3	manufacturing_max	manufacturing_sd
0	5	8	13.583695289007	23	50	11.555959199	0	4	6	7.0679362958013146	10	21	3.991366251182181

The descriptive statistics showed that both Manufacturing Lead Time and In-transit Lead Time varied substantially across shipments. The distributions also suggested the presence of variability associated with different shipment origins and shipping methods.

Categorical Variables

Frequency tables were calculated for the following categorical variables:

- LOB
- Origin
- Ship Mode
- Quarter
- Year

These summaries provided insight into the number of shipments associated with each category and helped identify dominant shipment patterns within the dataset.

Table 14 presents frequency summaries for the categorical variables included in the analysis.

Table 14. Categorical Variable Frequencies

variable	category	n
LOB	PRODUCT A	2901
LOB	PRODUCT B	5823
LOB	PRODUCT C	255
Origin	SITE A	3412
Origin	SITE B	1546

Origin	SITE C	1627
Origin	SITE D	2394
Ship Mode	AIR	3752
Ship Mode	FAST BOAT	523
Ship Mode	GROUND	2394
Ship Mode	OCEAN	2310
Quarter	Q1	3862
Quarter	Q2	3629
Quarter	Q3	1445
Quarter	Q4	43
Year	2019	43
Year	2020	8936

The categorical summaries showed that shipment activity varied across origins and ship modes, indicating that some transportation methods and shipment locations were used more frequently than others.

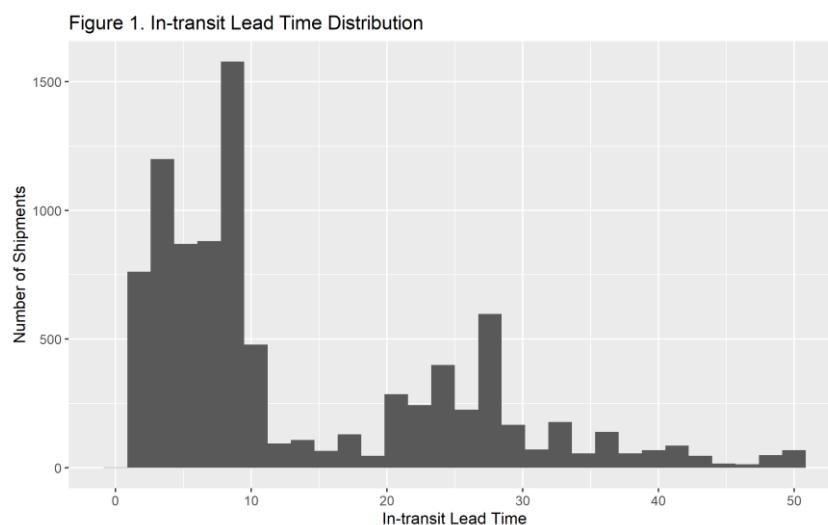
6. Graphical Exploration

Several graphical methods were used to explore the cleaned shipment data and examine relationships between variables.

Distribution of In-transit Lead Time

Figure 1 displays the distribution of In-transit Lead Time after cleaning. The histogram helped visualize the spread of shipment transportation times and identify the most common lead time ranges.

Figure 1. Distribution of In-transit Lead Time

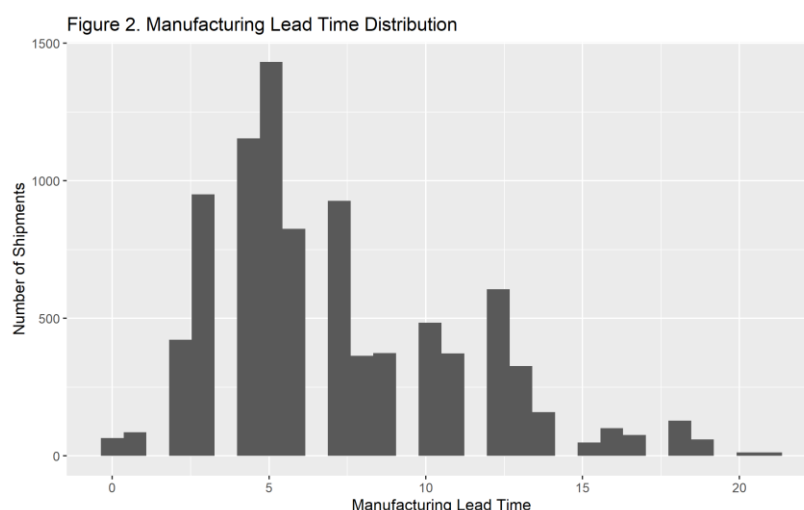


The distribution showed that most shipments were concentrated within a moderate range of transportation days, although some variation still existed across shipments.

Distribution of Manufacturing Lead Time

Figure 2 displays the distribution of Manufacturing Lead Time after cleaning.

Figure 2. Distribution of Manufacturing Lead Time

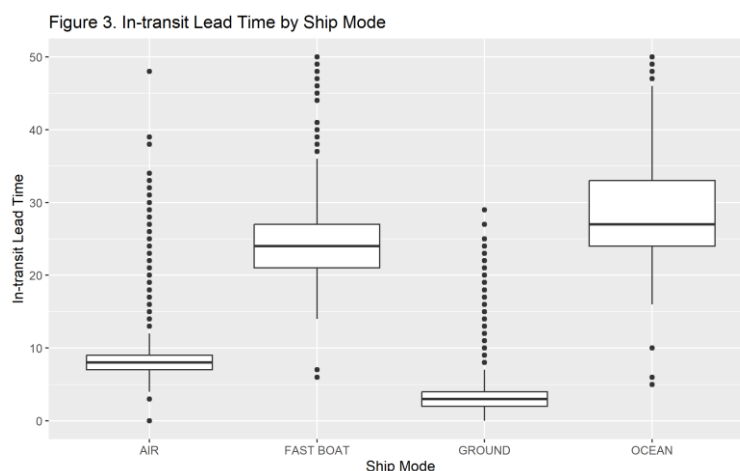


The distribution indicated variability in manufacturing preparation time before products were shipped from the manufacturing sites.

In-transit Lead Time by Ship Mode

Figure 3 compares In-transit Lead Time across the different shipping methods.

Figure 3. In-transit Lead Time by Ship Mode

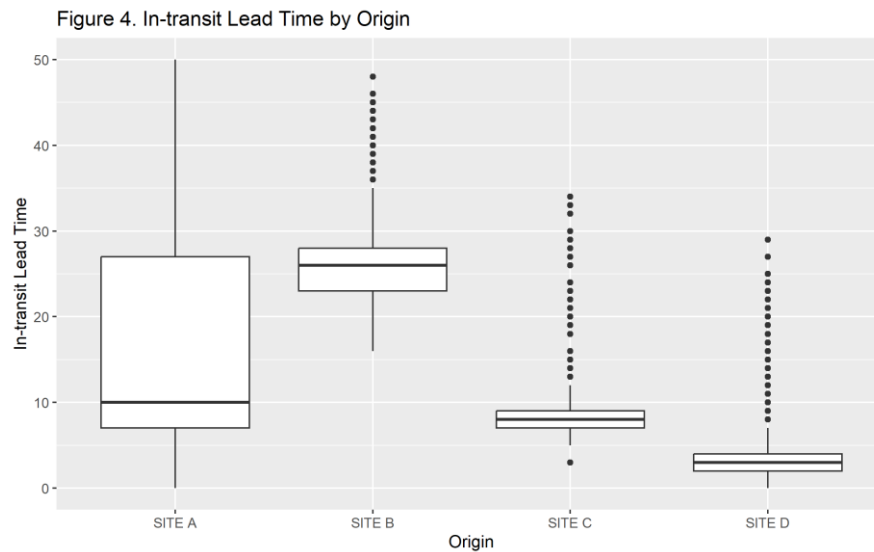


The boxplots showed noticeable differences in shipment time distributions among AIR, FAST BOAT, GROUND, and OCEAN shipping methods. Faster transportation methods generally produced lower lead times compared to slower shipping methods.

In-transit Lead Time by Origin

Figure 4 compares In-transit Lead Time across shipment origin sites.

Figure 4. In-transit Lead Time by Origin

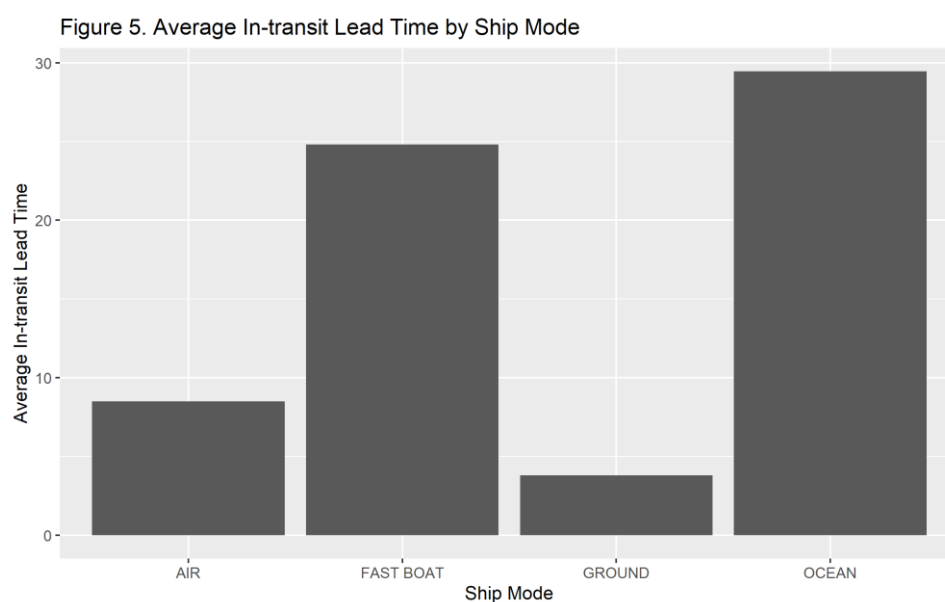


The figure suggested that some origin locations experienced longer and more variable shipment times than others.

Average In-transit Lead Time by Ship Mode

Figure 5 summarizes the average In-transit Lead Time for each shipping method.

Figure 5. Average In-transit Lead Time by Ship Mode

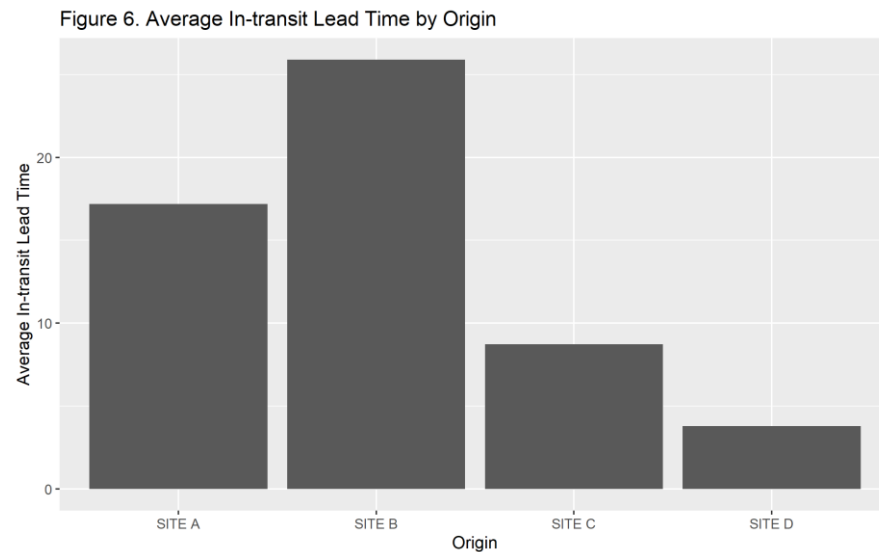


The results showed clear differences between transportation methods, with AIR shipments generally having lower average transportation times compared to OCEAN shipments.

Average In-transit Lead Time by Origin

Figure 6 presents the average In-transit Lead Time for each origin site.

Figure 6. Average In-transit Lead Time by Origin

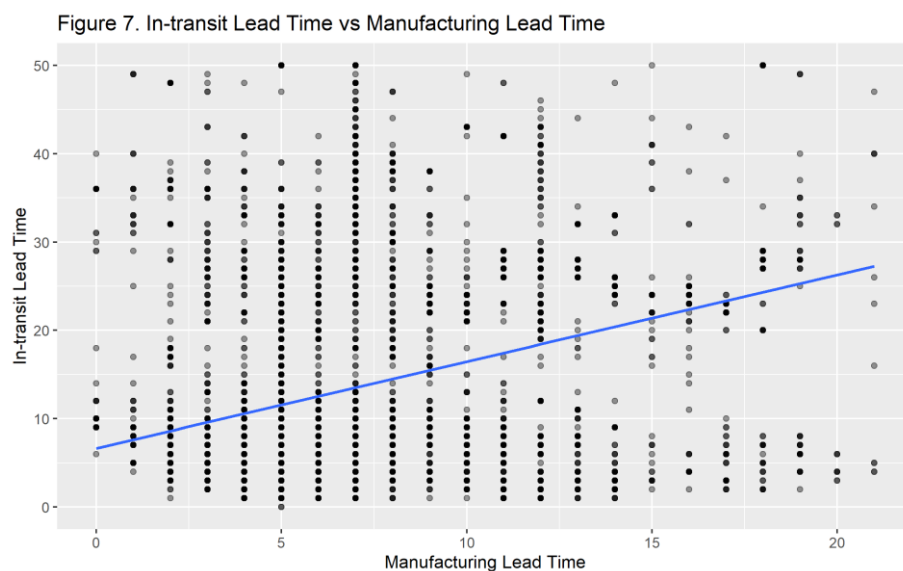


The figure demonstrated variation in average transportation time among the different shipment origins.

Relationship Between Manufacturing and In-transit Lead Time

Figure 7 displays the relationship between Manufacturing Lead Time and In-transit Lead Time.

Figure 7. Manufacturing Lead Time vs In-transit Lead Time

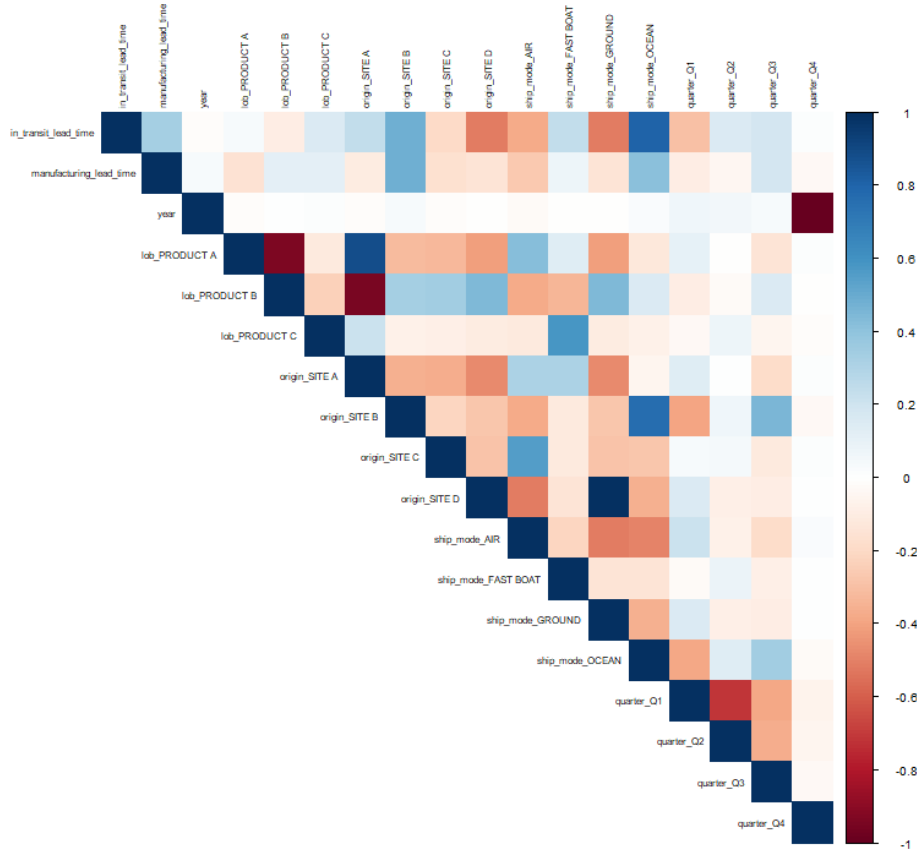


The scatterplot showed that only a limited linear relationship existed between manufacturing time and transportation time. Most shipment variability appeared to be associated with transportation and operational factors rather than manufacturing duration.

Correlation Plot

Figure 8 displays the full correlation plot for the numeric and one-hot encoded variables.

Figure 8. Correlation Plot



The correlation plot provided a visual summary of the relationships between variables and helped identify potential factors influencing shipment transportation performance.

7. Correlation Analysis

Correlation analysis was performed to examine relationships between In-transit Lead Time and the other predictor variables.

Because correlation analysis requires numeric variables, the categorical variables were converted into dummy variables using one-hot encoding. The categorical variables converted into dummy variables included:

- LOB
- Origin
- Ship Mode
- Quarter

A full correlation matrix was calculated using the cleaned numeric dataset.

Table 15 presents the average In-transit Lead Time by Ship Mode.

Table 15. Average In-transit Lead Time by Ship Mode

ship mode	avg in transit lead time
AIR	8.48774
FAST BOAT	24.803059273422562
GROUND	3.799916457811195
OCEAN	29.46017316017316

The table showed that shipment time performance varied substantially across transportation methods.

Table 16 presents the average In-transit Lead Time by Origin.

Table 16. Average In-transit Lead Time by Origin

origin	avg in transit lead time
SITE A	17.18493552168816
SITE B	25.891979301423028
SITE C	8.732022
SITE D	3.799916457811195

The results demonstrated that some manufacturing origins experienced longer transportation lead times than others.

Table 17 presents the complete correlation matrix for the variables included in the analysis.

Table 17. Full Correlation Matrix

in_transit_le	manufacturir	year	lob_PRODUC	lob_PROD	lob_PROD	origin_SITE	origin_SITE	origin_SITE	origin_SITE	ship_mode	ship_mode	ship_mode	ship_mode	quarter_Q	quarter_Q	quarter_Q	quarter_Q4
1	0.339310105	-0.0193918	0.0394813	-0.09266	0.1551591	0.2439855	0.4857785	-0.19752	-0.51052	-0.37364	0.2414653	-0.51052	0.8086265	-0.29184	0.1535235	0.1845313	0.0193917976482
0.339310105	1	0.03109161	-0.1588437	0.1155146	0.1151844	-0.10813	0.4871527	-0.16978	-0.14935	-0.2693	0.074759	-0.14935	0.4148750	-0.09437	-0.04001	0.1864136	-0.031091608
-0.019392	0.03109161	1	-0.0107188	0.006373	0.0118597	-0.01216	0.031636	-0.01344	-0.00195	-0.023	-0.00341	-0.00195	0.029755	0.060264	0.057132	0.03038	-1
0.0394813	-0.1588437	-0.0107188	1	-0.93842	-0.11812	0.8824693	-0.31508	-0.325	-0.41656	0.4228725	0.1362785	-0.41656	-0.12875	0.1083428	-0.00315	-0.14378	0.010718751
-0.09266	0.115514654	0.00637291	-0.9384225	1	-0.23223	-0.94038	0.3357513	0.3463268	0.4438940	-0.37327	-0.33781	0.4438940	0.1531202	-0.09308	-0.02398	0.1586364	-0.006372906
0.15515911	0.115184433	0.011859732	-0.1181152	-0.23223	1	0.2183827	-0.07797	-0.08043	-0.10309	-0.11766	0.5872620	-0.10309	-0.07761	-0.03748	0.077791	-0.05116	-0.011859733
0.24398559	-0.1081259	-0.0121646	0.88246932C	-0.94038	0.2183827	1	-0.35704	-0.36829	-0.47204	0.3173932	0.3176687	-0.47204	-0.05081	0.1313630	0.001865	-0.18176	0.0121646464763
0.48577859	0.487152751	0.03163627	-0.3150766	0.3357513	-0.07797	-0.35704	1	-0.21454	-0.27498	-0.37622	-0.11342	-0.27498	0.7634300	-0.39621	0.060811	0.4585461	-0.031636274
-0.197515	-0.1697763	-0.0134373	-0.3250009	0.3463268	-0.08043	-0.36829	-0.21454	1	-0.28365	0.5552465	-0.11699	-0.28365	-0.27686	0.0398312	0.040315	-0.11003	0.0134372616823
-0.510516	-0.1493464	-0.0019527	-0.4165601	0.4438940	-0.10309	-0.47204	-0.27498	-0.28365	1	-0.51085	-0.14995	0.9999999	-0.35486	0.1593830	-0.08908	-0.09614	0.001952711
-0.373636	-0.2693038	-0.0230003	0.422872591	-0.37327	-0.11766	0.3173932	-0.37622	0.5552465	-0.51085	1	-0.2107	-0.51085	-0.49863	0.2117242	-0.0798	-0.18301	0.023000338
0.24146532	0.0747587	-0.0034121	0.13627855	-0.33781	0.5872620	0.3176687	-0.11342	-0.11699	-0.14995	-0.2107	1	-0.14995	-0.14637	-0.02396	0.083936	-0.08045	0.003412131
-0.510516	-0.1493464	-0.0019527	-0.4165601	0.4438940	-0.10309	-0.47204	-0.27498	-0.28365	0.9999999	-0.51085	-0.14995	1	-0.35486	0.1593830	-0.08908	-0.09614	0.001952711
0.80862651	0.414875014	0.02975462	-0.1287535	0.1531202	-0.07761	-0.05081	0.7634300	-0.27686	-0.35486	-0.49863	-0.14637	-0.35486	1	-0.38728	0.1351841	0.3468417	-0.029754621
-0.291838	-0.0943714	0.06026443	0.108342804	-0.09308	-0.03748	0.1313630	-0.39621	0.0398312	0.1593830	0.2117242	-0.02396	0.1593830	-0.38728	1	-0.71551	-0.38047	-0.060264433
0.15352359	-0.0400057	0.05713198	-0.0031464	-0.02398	0.077791	0.001865	0.060811	0.040315	-0.08908	-0.0798	0.083936	-0.08908	0.1351841	-0.71551	1	-0.36069	-0.057131975
0.18453134	0.186413627	0.03037972	-0.1437832	0.1586364	-0.05116	-0.18176	0.4585461	-0.11003	-0.09614	-0.18301	-0.08045	-0.09614	0.3468417	-0.38047	-0.36069	1	-0.03037972
0.01939179	-0.0310916	-1	0.0107188	-0.00637	-0.01186	0.0121646	-0.03164	0.0134372	0.001953	0.023	0.003412	0.001953	-0.02975	-0.06026	-0.05713	-0.03038	1

The correlation matrix helped identify positive and negative relationships among the variables used in the analysis.

Table 18 presents the correlations between In-transit Lead Time and the remaining variables.

Table 18. Correlation With In-transit Lead Time

predictor	in transit lead time	abs correlation	
ship_mode OCEAN	0.8086265173044068	0.8086265173044068	
origin SITE D	-0.51052	0.5105155740468049	
ship_mode GROUND	-0.51052	0.5105155740468049	
origin SITE B	0.4857785931589096	0.4857785931589096	
ship_mode AIR	-0.37364	0.37363622024850973	
manufacturing lead time	0.3393101052518675	0.3393101052518675	
quarter Q1	-0.29184	0.29183786833149744	
origin SITE A	0.24398559119088636	0.24398559119088636	
ship_mode_FAST BOAT	0.2414653287402479	0.2414653287402479	
origin SITE C	-0.19752	0.19751530294677597	
quarter Q3	0.18453134364193832	0.18453134364193832	
lob_PRODUCT C	0.15515911071000765	0.15515911071000765	
quarter Q2	0.15352359533306464	0.15352359533306464	
lob_PRODUCT B	-0.09266	0.092656	
lob_PRODUCT A	0.039481	0.039481	
year	-0.01939	0.019391797648203483	
quarter Q4	0.019391797648203483	0.019391797648203483	

The correlation analysis identified which variables had the strongest associations with In-transit Lead Time. Ship Mode variables demonstrated stronger relationships with transportation lead time than many of the other variables in the dataset.

8. Conclusion

This project successfully prepared and explored ABC's shipment dataset using R programming techniques for data science. The analysis imported and combined data from the Raw-Data and Calendar worksheets, created Quarter and Year variables, calculated Manufacturing Lead Time and In-transit Lead Time, cleaned missing and unusual values, and explored the cleaned dataset using summary statistics, graphs, and correlation analysis.

The data cleaning process standardized categorical variables, removed problematic observations, and imputed missing lead time values to improve overall data quality. Exploratory analysis showed that shipment transportation times varied across shipping methods and origin locations, indicating that operational factors may significantly affect logistics performance.

The final cleaned dataset provides a reliable foundation for future predictive modeling and supply chain analysis. ABC can use this prepared dataset to further investigate the impact of shipping methods, manufacturing sites, and operational decisions on logistics lead time performance and customer delivery expectations.